FERTILITY ASSESSMENT AND INVESTIGATION MANAGEMENT SYSTEM USING PYTHON OOP

1. Background and Problem Statement

Fertility assessment may involve several consultations and investigations, including reproductive history, laboratory tests, imaging and assessment of the male partner. The American Society for Reproductive Medicine recommends a systematic evaluation to identify factors that may affect conception [1]. However, when records are maintained across different visits, missing results and poorly documented follow-ups can make it difficult to track completed and pending investigations, potentially delaying clinical review, diagnosis and management [2].

This project will develop a Python-based **Fertility Assessment and Investigation Management System** to organize patient profiles, consultations and investigations. It will validate records, flag missing results, track follow-ups and generate assessment summaries. The system will support organized and consistent management of fertility assessment records but will not diagnose infertility or recommend treatment.

2. Aim: To develop and test an object-oriented Python application for managing fertility assessment records.

2.1 Objectives:

- To design and implement Python classes for managing fertility assessment records, including patient profiles, consultations, appointments and investigations.
- To apply encapsulation, inheritance, polymorphism and abstraction in developing the fertility assessment system.
- To develop data validation functions for identifying missing investigation results, duplicate records, invalid dates and inconsistent investigation statuses.
- To develop a simple user interface for recording assessments, tracking investigation progress and monitoring follow-up appointments.

- To test the system using fictional fertility assessment records to evaluate its functionality, data validation and accuracy of generated summaries.

3. System Design and OOP Implementation

The application will use Python for core functionality, SQLite for data storage and Streamlit for the user interface. Pandas may support data processing.

The main classes will include **Woman**, **FertilityAssessment**, **AssessmentRecord**, **Investigation**, **Appointment**, **DataQualityChecker** and **ReportGenerator**. Investigation and Appointment will inherit shared attributes and methods from the abstract AssessmentRecord class while implementing their own summary methods. This will demonstrate inheritance and polymorphism. Encapsulation will be applied through controlled methods for creating and updating records, while abstraction will separate shared record behaviour from specific implementation details.

The system will link each patient to the relevant assessment, consultations and investigations. It will validate entered information, identify missing or inconsistent records, and generate summaries of investigation status and outstanding follow-ups. Exception handling will help manage invalid inputs and database errors.

4. Scope, Methodology and Testing

Development will follow an incremental approach over three weeks. The system will use approximately 50 fictional records representing different consultation histories, investigation statuses and data-quality issues. No real patient data will be used.

Testing will cover class methods, object relationships, validation rules, exception handling, database operations and report accuracy. System outputs will be compared with expected results from predefined test cases.

5. Expected Deliverables: Deliverables will include a working Python application, SQLite database, fictional dataset, modular and documented OOP code, automated

tests, an interactive interface, assessment summaries and a GitHub repository containing the source code and README instructions.

References

[1] Practice Committee of the American Society for Reproductive Medicine, “Fertility evaluation of infertile women: A committee opinion,” *Fertility and Sterility*, vol. 116, no. 5, pp. 1255–1265, Nov. 2021, doi: 10.1016/j.fertnstert.2021.08.038.

[2] American College of Obstetricians and Gynecologists, “Evaluating infertility,” Apr. 2020. [Online]. Available: https://www.acog.org/womens-health/faqs/evaluating-infertility. [Accessed: Oct. 11, 2026].